FP2 Chapter 13 Complex Numbers

Lecture 13 Complex Numbers

(De Moivre's Theorem, Trigonometry Function, Binomial Expansions, Roots of Unity, Summation)

De Moivre's Theorem

- The theorem states that $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.
- To prove that it is valid for **positive integer** values of n, we use the method of induction (from PYQ O/N 2006 Q11).

Proof:

When $n = 1$, LHS $= (\cos\theta + i\sin\theta)^1$

$= \text{RHS}$

Hence it is true for $n = 1$.

Assume that it is true for $n = k$, $(\cos\theta + i\sin\theta)^k = \cos k\theta + i\sin k\theta$

For $n = k + 1$, $\quad (\cos\theta + i\sin\theta)^{k+1}$

$$= (\cos\theta + i\sin\theta)^k(\cos\theta + i\sin\theta)$$

$$= (\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)$$

$$= \cos k\theta\cos\theta - \sin k\theta\sin\theta + i(\sin k\theta\cos\theta + \cos k\theta\sin\theta)$$

$$= \cos(k+1)\theta + i\sin(k+1)\theta$$

Hence it is true for $n = k + 1$.

By induction, it is true for $n = 1,2,3, \ldots$

- For **negative integers**, we show that if it is true for n, it is also true for $-n$.

Proof:

$$(\cos\theta + i\sin\theta)^{-n} = \frac{1}{(\cos\theta + i\sin\theta)^n} = \frac{1}{\cos n\theta + i\sin n\theta} = \frac{\cos n\theta - i\sin n\theta}{\cos^2 n\theta + \sin^2 n\theta}$$

$$= \cos n\theta - i\sin n\theta = \cos(-n)\theta + i\sin(-n)\theta$$

Note:

When n rational fraction, $(\cos\theta + i\sin\theta)^{\frac{p}{q}} = \cos\frac{p}{q}\theta + i\sin\frac{p}{q}\theta$.

$\cos\frac{p}{q}\theta + i\sin\frac{p}{q}\theta$ is just one value of $(\cos\theta + i\sin\theta)^{\frac{p}{q}}$. There are further values, as subsequent work will show.

Applications of De Moivre's Theorem:

(I) Express trigonometrical ratios of multiple angles in terms of powers of trigonometrical ratios of the fundamental angle.

- Strategy: Expand RHS using binomial theorem then equate real or imaginary part.
- Main purpose: To solve polynomial equations.

$\cos n\theta + i\sin n\theta = (c + is)^n$, where $c = \cos\theta, s = \sin\theta$

$$= \binom{n}{0}c^n + \binom{n}{1}c^{n-1}(is)^1 + \binom{n}{2}c^{n-2}(is)^2 + \binom{n}{3}c^{n-3}(is)^3 + \binom{n}{4}c^{n-4}(is)^4 + \binom{n}{5}c^{n-5}(is)^5 \ldots + \binom{n}{n}(is)^n$$

$\cos(n\theta) = \mathrm{Re}(c + is)^n$

$$= \binom{n}{0}c^n - \binom{n}{2}c^{n-2}s^2 + \binom{n}{4}c^{n-4}s^4 - \cdots$$

$\sin(n\theta) = \mathrm{Im}(c + is)^n$

$$= \binom{n}{1}c^{n-1}s - \binom{n}{3}c^{n-2}s^3 + \binom{n}{5}c^{n-5}s^5 - \cdots$$

- **Example 1 (PYQ M/J 2011/11):**

Use de Moivre's theorem to prove that

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}. \qquad [6]$$

State the exact values of θ, between 0 and π, that satisfy $\tan 3\theta = 1$. [2]

Express each root of the equation $t^3 - 3t^2 - 3t + 1 = 0$ in the form $\tan(k\pi)$, where k is a positive rational number. [3]

For each of these values of k, find the exact value of $\tan(k\pi)$. [3]

- **Example 2 (PYQ O/N 2012/11):**

Use de Moivre's theorem to show that

$$\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1.$$ [3]

Without using a calculator, verify that $\cos 4\theta = -\cos 3\theta$ for each of the values $\theta = \frac{1}{7}\pi, \frac{3}{7}\pi, \frac{5}{7}\pi, \pi$. [2]

Using the result $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, show that the roots of the equation

$$8c^4 + 4c^3 - 8c^2 - 3c + 1 = 0$$

are $\cos\frac{1}{7}\pi, \cos\frac{3}{7}\pi, \cos\frac{5}{7}\pi, -1$. [2]

Deduce that $\cos\frac{1}{7}\pi + \cos\frac{3}{7}\pi + \cos\frac{5}{7}\pi = \frac{1}{2}$. [2]

- **Example 3 (PYQ O/N 2010):**

By using de Moivre's theorem to express $\sin 5\theta$ and $\cos 5\theta$ in terms of $\sin\theta$ and $\cos\theta$, show that

$$\tan 5\theta = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4},$$

where $t = \tan\theta$. [5]

Show that the roots of the equation $x^4 - 10x^2 + 5 = 0$ are $\tan\left(\frac{1}{5}n\pi\right)$ for $n = 1, 2, 3, 4$. [2]

By considering the product of the roots of this equation, find the exact value of $\tan\left(\frac{1}{5}\pi\right)\tan\left(\frac{2}{5}\pi\right)$. [3]

- **Example 4:**

Use de Moivre's theorem to show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$.

By considering the equation $\cos 5\theta = 0$, prove that $\cos\left(\frac{\pi}{10}\right)\cos\left(\frac{3\pi}{10}\right) = \frac{1}{4}\sqrt{5}$.

- **Example 5 (PYQ M/J 2010/13):**

Use de Moivre's theorem to show that

$$\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta. \quad [4]$$

Hence find all the roots of the equation

$$32x^5 - 40x^3 + 10x + 1 = 0$$

in the form $\sin(q\pi)$, where q is a positive rational number. [4]

- **Example 6:**

Use de Moivre's theorem to express $\cos 6\theta$ in terms of $\cos\theta$ and $\sin\theta$, and express $\sin 6\theta$ in a similar form.

Deduce that

$$\tan 6\theta = \frac{6t - 20t^3 + 6t^5}{1 - 15t^2 + 15t^4 - t^6}, \text{where } t = \tan\theta.$$

By giving θ a suitable value, prove that one of the roots of the cubic equation $x^3 - 15x^2 + 15x - 1 = 0$ is $\tan^2\left(\frac{\pi}{12}\right)$. State the other two roots of the equation. Deduce that $\tan^2\left(\frac{\pi}{12}\right) = 7 - 4\sqrt{3}$.

- **Example 7:**

Express $\tan 7\theta$ as quotient of two polynomials in t, where $t = \tan\theta$.

Show that if $\theta = \frac{\pi}{7}$, then $t^6 - 21t^4 + 35t^2 - 7 = 0$.

Deduce the numerical value of $\sec^2\left(\frac{\pi}{7}\right) + \sec^2\left(\frac{2\pi}{7}\right) + \sec^2\left(\frac{3\pi}{7}\right)$.

(II) Express powers of $\sin\theta$ and $\cos\theta$ in terms of multiple angles.

- Main purpose: Integrating trigonometric functions (an alternative to reduction formulae)

Properties of z and $\frac{1}{z}$:

$$z = \cos\theta + i\sin\theta$$

$$z^n = \cos n\theta + i\sin n\theta \quad (1)$$

$$\frac{1}{z^n} = z^{-n} = \cos(-n)\theta + i\sin(-n)\theta = \cos n\theta - i\sin n\theta \qquad (2)$$

$$(1)+(2): \quad z^n + \frac{1}{z^n} = 2\cos n\theta = e^{in\theta} + e^{-in\theta}$$

$$(1)-(2): \quad z^n - \frac{1}{z^n} = 2i\sin n\theta = e^{in\theta} - e^{-in\theta}$$

- The two results above are used to express $\sin^n\theta$ and $\cos^n\theta$ in terms of multiple angles of sine and cosine.

- Strategy: Expand using binomial theorem, group the terms and use the two results above.

- **Example 8:**

  Show that if $z = \cos\theta + i\sin\theta$, then $z^n - \frac{1}{z^n} = 2i\sin n\theta$, where n is a positive integer. Hence, or otherwise, show that $\sin^7\theta$ can be expressed in the form $a\sin\theta + b\sin 3\theta + c\sin 5\theta + d\sin 7\theta$, where a, b, c, d are constants to be determined.

- **Example 9:**

Show that $\cos^3\theta\sin\theta = \frac{1}{8}(\sin 4\theta + 2\sin 2\theta)$.

- **Example 10:**

Use de Moivre's theorem to prove that $\cos^6\theta + \sin^6\theta = \frac{1}{8}(3\cos 4\theta + 5)$.

- **Example 11:**

If $z - \frac{1}{z} = i$, find the value of $z^7 - \frac{1}{z^7}$.

- **Example 12:**

If $z = \cos\theta + i\sin\theta$ where $\sin\frac{1}{2}\theta \neq 0$, prove that $\frac{z+1}{z-1} = -\frac{1}{2}\cot\frac{1}{2}\theta$.

- **Example 13 (PYQ M/J 2012/13):**

7 Expand $\left(z+\frac{1}{z}\right)^4\left(z-\frac{1}{z}\right)^2$ and, by substituting $z=\cos\theta+\mathrm{i}\sin\theta$, find integers p, q, r, s such that

$$64\sin^2\theta\cos^4\theta = p + q\cos 2\theta + r\cos 4\theta + s\cos 6\theta.$$ [6]

Using the substitution $x = 2\cos\theta$, show that

$$\int_1^2 x^4\sqrt{(4-x^2)}\,\mathrm{d}x = \tfrac{4}{3}\pi + \sqrt{3}.$$ [4]

- **Example 14:**

Use de Moivre's theorem to express $\sin^4\theta$ and $\sin^6\theta$ in terms of multiple angles of sine and cosine and prove that

$$\int_0^{\frac{\pi}{6}} \sin^4\theta\cos 2\theta\, d\theta = \frac{5\sqrt{3}}{64} - \frac{\pi}{24}.$$

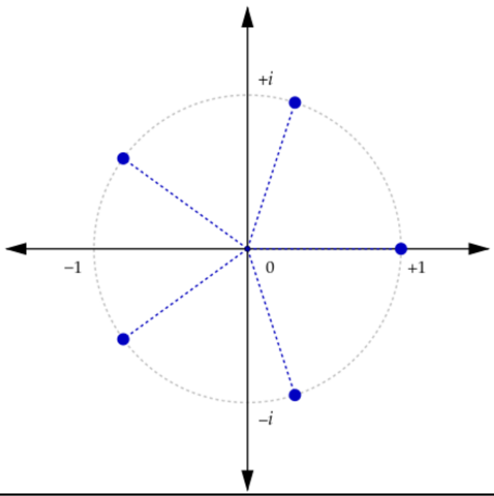
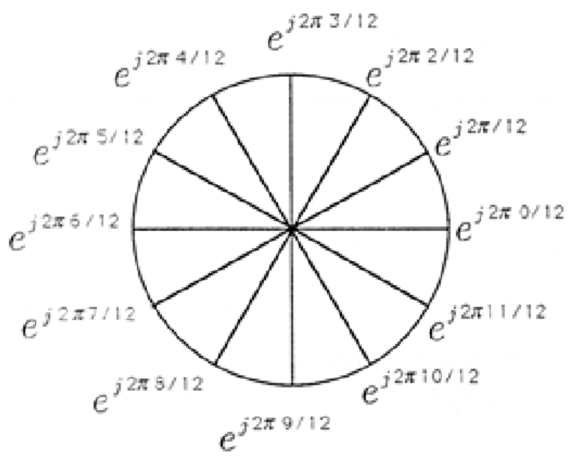
(III) To find the nth roots of unity of any complex number.

Case 1: To find the nth roots of unity

- Means the nth roots of 1.

- For example, cube roots of 1 are $1, \frac{1}{2}+\frac{\sqrt{3}}{2}i, \frac{1}{2}-\frac{\sqrt{3}}{2}i$. The roots can be obtained by solving $x^3-1=0$.

- If the nth roots are plotted in an Argand diagram, the lines will form a circle and are $\frac{2k\pi}{n}$ away from each other as seen below.

5^{th} roots of unity	**12^{th} roots of unity**
 	
Angle between each root is $\frac{2}{5}\pi$	Angle between each root is $\frac{2}{12}\pi$

- Derivation of the nth roots of unity:

Let $z=1^{\frac{1}{n}}=r(\cos\theta+i\sin\theta)$

Raise both sides to the nth power:

$$z^n=r^n(\cos n\theta+i\sin n\theta)=1=\cos 0+i\sin 0$$

Comparing the coefficients: $r^n=1\rightarrow r=1$ since r is real.

$n\theta=0,2\pi,4\pi,\ldots 2(n-1)\pi$, since we want n values of θ,

$\theta=\frac{2k\pi}{n}$, where $k=0,1,2,..,n-1$

$$z = \cos\left(\frac{2k\pi}{n}\right) + i\sin\left(\frac{2k\pi}{n}\right) = e^{i\left(\frac{2k\pi}{n}\right)}$$

- This method can be extended to find roots of other numbers (you can use this to do square root problem in P3 when $n = 2$).

- **Note: Special Conjugate method**

- **Example 15:**

Find, in any form, the 8 complex numbers which satisfy the equation $z^8 - 1 = 0$, and represent them in Argand diagram.

Use a graphical argument to find the greatest value and the least value of $|z - 5 - 5i|$.

- **Example 16:**

The complex number ω, where $\omega \neq 1$, is a root of the equation $z^3 - 1 = 0$.

Show that $1 + \omega + \omega^2 = 0$.

Hence show that $1 + \omega^k + \omega^{2k}$, where k is a positive integer, is not zero when k is a multiple of 3, but is zero when k is not a multiple of 3.

- **Example 17:**

Find, in any form, all the roots of the equation $z^5 - 1 = 0$.

Hence show that the 5 complex numbers which satisfy the equation $\left(\frac{2\omega+1}{\omega}\right)^5 = 1$ are

$$\frac{-2 + \cos\left(\frac{2k\pi}{5}\right) - i\sin\left(\frac{2k\pi}{5}\right)}{5 - 4\cos\left(\frac{2k\pi}{5}\right)}, \text{where } k = 0,1,2,3,4.$$

- **Example 18 (PYQ O/N 2007):**

9 Write down, in any form, all the roots of the equation $z^5 - 1 = 0$. [2]

Hence find all the roots of the equation

$$(w-1)^4 + (w-1)^3 + (w-1)^2 + w = 0,$$

and deduce that none of them is real. [4]

Find the arguments of the two roots which have the smaller modulus. [4]

- **Example 19 (PYQ M/J 2010/12):**

9 **(i)** Write down the five fifth roots of unity. [2]

(ii) Hence find all the roots of the equation

$$z^5 + 16 + (16\sqrt{3})\mathrm{i} = 0,$$

giving answers in the form $r\mathrm{e}^{\mathrm{i}q\pi}$, where $r > 0$ and q is a rational number. Show these roots on an Argand diagram. [4]

Let w be a root of the equation in part **(ii)**.

(iii) Show that

$$\sum_{k=0}^{4}\left(\frac{w}{2}\right)^k = \frac{3 + \mathrm{i}\sqrt{3}}{2 - w}.$$ [3]

(iv) Identify the root for which $|2 - w|$ is least. [2]

Case 2: To find the factors of $z^n - 1$

- When we are finding roots of 1, we know that they all lie on the unit circle and that they are equally spread around the circle.

- When we are finding the nth roots of 1, we are solving the equation $z^n - 1 = 0$.

$$z^n - 1 = 0$$

$$z^n = 1$$

$$z = 1^{\frac{1}{n}}$$

$$z = (\cos 2k\pi + i\sin 2k\pi)^{\frac{1}{n}} = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n} = e^{i\frac{2k\pi}{n}}$$

where $k = 0,1,2, \dots n-1$ or $k = 0, \pm 1, \pm 2, \dots$

- You can use the results $\left(z - e^{in\theta}\right)\left(z - e^{-in\theta}\right) = z^2 - (2\cos\theta)z + 1$ to factorise $z^n - 1$.

- You will remember from P3 that, since $z^n - 1 = 0$ is just a polynomial equation of degree n with real coefficients, the complex roots must occur in conjugate pairs.

- In addition, from FP1 syllabus, this is just a polynomial and so we can find the sum of the roots. Obviously, the sum of the roots is zero.

- One of these roots is 1, the others are the complex numbers of the form $e^{i\frac{2k\pi}{n}}$, where $k = 0,1,2,..,n-1$.

Notice:

- When n is **even**, there must be at least one more real root. In fact, this is $-1\ (= e^{i\pi})$.

- When n is **odd**, then 1 is the only real root and the other form complex conjugate pairs.

- **Example 20:**

Find all the solutions to the equations

(i) $z^3 - 1 = 0.$

(ii) $z^6 - 1 = 0.$

And factorise them.

- **Example 21 (PYQ O/N 2004):**

6 Write down all the 8th roots of unity. [2]

Verify that

$$\left(z - e^{i\theta}\right)\left(z - e^{-i\theta}\right) \equiv z^2 - (2\cos\theta)z + 1. \quad [1]$$

Hence express $z^8 - 1$ as the product of two linear factors and three quadratic factors, where all coefficients are real and expressed in a non-trigonometric form. [5]

- **Example 22 (PYQ O/N 2011/11):**

Let $\omega = \cos\frac{1}{5}\pi + \mathrm{i}\sin\frac{1}{5}\pi$. Show that $\omega^5 + 1 = 0$ and deduce that

$$\omega^4 - \omega^3 + \omega^2 - \omega = -1.$$ [2]

Show further that

$$\omega - \omega^4 = 2\cos\tfrac{1}{5}\pi \quad \text{and} \quad \omega^3 - \omega^2 = 2\cos\tfrac{3}{5}\pi.$$ [4]

Hence find the values of

$$\cos\tfrac{1}{5}\pi + \cos\tfrac{3}{5}\pi \quad \text{and} \quad \cos\tfrac{1}{5}\pi\cos\tfrac{3}{5}\pi.$$ [4]

Find a quadratic equation having roots $\cos\frac{1}{5}\pi$ and $\cos\frac{3}{5}\pi$ and deduce the exact value of $\cos\frac{1}{5}\pi$. [4]

- **Example 23:**

(a) Find all the solutions to the equation $z^9 - 1 = 0$.

(b) Show that

$$\sin\frac{2\pi}{9} + \sin\frac{4\pi}{9} + \sin\frac{2\pi}{3} + \cdots + \sin\frac{16\pi}{9} = 0$$

and

$$\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} + \cos\frac{2\pi}{3} + \cos\frac{8\pi}{9} = -\frac{1}{2}.$$

Case 3: To find the nth roots of a complex number

Let $z^n = r\,(\cos\theta + i\sin\theta)$ where $n \in Z^+$.

$$z^n = r[\cos(2k\pi + \theta) + i\sin(2k\pi + \theta)] \text{ where } k \in Z$$

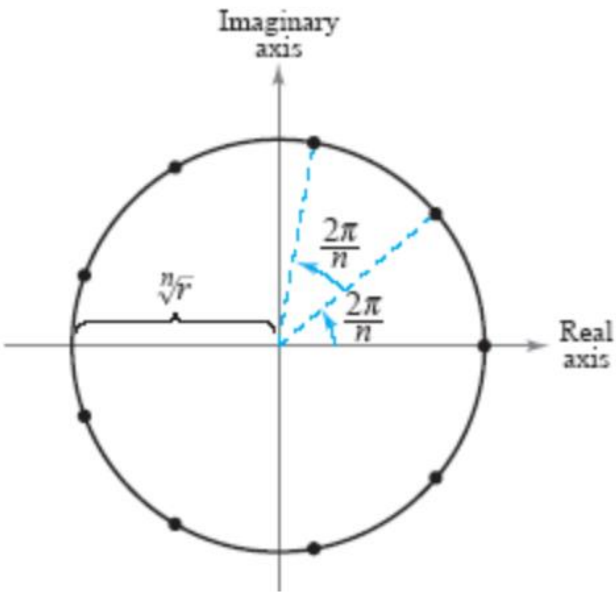
$$z = r^{\frac{1}{n}}[\cos(2k\pi + \theta) + i\sin(2k\pi + \theta)]^{\frac{1}{n}}$$

$$z = r^{\frac{1}{n}}\left[\cos\left(\frac{2k\pi + \theta}{n}\right) + i\sin\left(\frac{2k\pi + \theta}{n}\right)\right], \quad k = 0,1,2,\dots,n-1$$

- There are nth distinct roots.

Notice:

- The modulus of each root is $\sqrt[n]{r}$.
- The argument starts with $\frac{\theta}{n}$ and increase by $\frac{2\pi}{n}$ for the consecutive roots.



- **Example 24 (PYQ 9709 O/N 2002):**

  Find two square roots of the complex number $-3+4i$, giving your answers in the form $x+iy$, where x and y are real.

- **Example 25:**

Express $8(\sqrt{3} - i)$ in the form $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$, giving θ in terms of π.

Hence obtain the roots of the equation $z^4 = 8(\sqrt{3} - i)$ in the same form.

- **Example 26 (PYQ O/N 2005):**

1 Write down the fifth roots of unity. Hence, or otherwise, find all the roots of the equation

$$z^5 = -16 + (16\sqrt{3})\mathrm{i},$$

giving each root in the form $r\mathrm{e}^{\mathrm{i}\theta}$. [4]

- **Example 27 (PYQ M/J 2004):**

7 Find the roots of the equation

$$z^3 = -(4\sqrt{3}) + 4\mathrm{i},$$

giving your answers in the form $r\mathrm{e}^{\mathrm{i}\theta}$, where $r > 0$ and $0 \leqslant \theta < 2\pi$. [5]

Denoting these roots by z_1, z_2, z_3, show that, for every positive integer k,

$$z_1^{3k} + z_2^{3k} + z_3^{3k} = 3\left(2^{3k}\mathrm{e}^{\frac{5}{6}k\pi\mathrm{i}}\right).$$ [4]

(IV) To find the sum of the first n terms of trigonometrical series.

$$\sum_{k=1}^{n} a^{g(k)} \cos f(k)\,\theta = \mathrm{Re} \sum_{k=1}^{n} a^{g(k)} z^{f(k)} \ \text{ or } \ \mathrm{Re} \sum_{k=1}^{n} a^{g(k)} e^{if(k)},$$

where $z = \cos\theta + i\sin\theta = e^{i\theta}$,

$$\sum_{k=1}^{n} a^{g(k)} \sin f(k)\,\theta = \mathrm{Im} \sum_{k=1}^{n} a^{g(k)} z^{f(k)} \ \text{ or } \ \mathrm{Im} \sum_{k=1}^{n} a^{g(k)} e^{if(k)},$$

where $z = \cos\theta + i\sin\theta = e^{i\theta}$,

where $g(k)$ and $f(k)$ are linear functions.

Note that $\sum_{k=1}^{n} a^{g(k)} z^{f(k)}$ is the sum of the first n terms of a geometric series.

- Main strategy for this question:

1. Expand one side using de Moivre's theorem.
2. Then expand another side using geometric series or binomial theorem.
3. Then equate the real part (cosine) or imaginary part (sine) for the complete proof.
4. You can also differentiate / integrate both sides of the equation or substitute special values to prove specific cases.

- Factor Formula:

$$\sin P + \sin Q = 2\sin\left(\frac{P+Q}{2}\right)\cos\left(\frac{P-Q}{2}\right)$$

$$\sin P - \sin Q = 2\cos\left(\frac{P+Q}{2}\right)\sin\left(\frac{P-Q}{2}\right)$$

$$\cos P + \cos Q = 2\cos\left(\frac{P+Q}{2}\right)\cos\left(\frac{P-Q}{2}\right)$$

$$\cos P - \cos Q = -2\sin\left(\frac{P+Q}{2}\right)\sin\left(\frac{P-Q}{2}\right)$$

- **Example 28:**

Show by using de Moivre's theorem, or otherwise, that provided $\cos\theta \neq 0$,

$$\sum_{k=1}^{10}(-1)^{k-1}\cos(2k-1)\theta = \frac{\sin^2 10\theta}{\cos\theta}.$$

- **Example 29:**

Show that, provided θ is not a multiple of 2π,

$$\sum_{k=0}^{6} \cos\theta = \cos 3\theta \sin\frac{7}{2}\theta \operatorname{cosec}\frac{1}{2}\theta.$$

- **Example 30:**

Show that, for $z \neq 2$,

$$\sum_{k=0}^{n} \left(\frac{z}{2}\right)^k = \frac{2^{n+1} - z^{n+1}}{2^n(2 - z)},$$

and deduce by taking $z = e^{i\theta}$ that

$$\sum_{k=0}^{n} \frac{\cos k\theta}{2^k} = \frac{2^{n+2} - 2^{n+1}\cos\theta + \cos n\theta - 2\cos(n+1)\theta}{2^n(5 - 4\cos\theta)}.$$

- **Example 31 (PYQ M/J 2015/11):**

8 By considering $\sum_{r=1}^{n} z^{2r-1}$, where $z = \cos\theta + \mathrm{i}\sin\theta$, show that, if $\sin\theta \neq 0$,

$$\sum_{r=1}^{n} \sin(2r-1)\theta = \frac{\sin^2 n\theta}{\sin\theta}. \qquad [7]$$

Deduce that

$$\sum_{r=1}^{n} (2r-1)\cos\left[\frac{(2r-1)\pi}{2n}\right] = -\operatorname{cosec}\left(\frac{\pi}{2n}\right)\cot\left(\frac{\pi}{2n}\right). \qquad [4]$$

- Example 32 (PYQ M/J 2014/13):

5 State the sum of the series $z + z^2 + z^3 + \ldots + z^n$, for $z \neq 1$. [1]

By letting $z = \cos\theta + \mathrm{i}\sin\theta$, show that

$$\cos\theta + \cos 2\theta + \cos 3\theta + \ldots + \cos n\theta = \frac{\sin\frac{1}{2}n\theta}{\sin\frac{1}{2}\theta}\cos\tfrac{1}{2}(n+1)\theta,$$

where $\sin\frac{1}{2}\theta \neq 0$. [7]

- **Example 33 (PYQ M/J 2009/12):**

By considering $\sum_{k=0}^{n-1}(1+\mathrm{i}\tan\theta)^k$, show that

$$\sum_{k=0}^{n-1}\cos k\theta\sec^k\theta=\cot\theta\sin n\theta\sec^n\theta,$$

provided θ is not an integer multiple of $\frac{1}{2}\pi$. [7]

Hence or otherwise show that

$$\sum_{k=0}^{n-1}2^k\cos\left(\tfrac{1}{3}k\pi\right)=\frac{2^n}{\sqrt{3}}\sin\left(\tfrac{1}{3}n\pi\right).$$ [2]

Given that $0<x<1$, show that

$$\sum_{k=0}^{n-1}\frac{\cos(k\cos^{-1}x)}{x^k}=\frac{\sin(n\cos^{-1}x)}{x^{n-1}\sqrt{(1-x^2)}}.$$ [4]

Using Binomial Series:

$$1+\binom{n}{1}\cos k\theta+\binom{n}{2}\cos 2k\theta+\binom{n}{3}\cos 3k\theta+\cdots+\binom{n}{n}\cos kn\theta=\mathrm{Re}(1+z^k)^n$$

$$\binom{n}{1}\sin k\theta+\binom{n}{2}\sin 2k\theta+\binom{n}{3}\sin 3k\theta+\cdots+\binom{n}{n}\sin kn\theta=\mathrm{Im}(1+z^k)^n$$

- **Example 34 (PYQ O/N 2012/13):**

8 Let $z=\cos\theta+\mathrm{i}\sin\theta$. Show that

$$1+z=2\cos\tfrac{1}{2}\theta(\cos\tfrac{1}{2}\theta+\mathrm{i}\sin\tfrac{1}{2}\theta).$$ [3]

By considering $(1+z)^n$, where n is a positive integer, deduce the sum of the series

$$\binom{n}{1}\sin\theta+\binom{n}{2}\sin 2\theta+\ \dots\ +\binom{n}{n}\sin n\theta.$$ [6]

- **Example 35 (PYQ M/J 2015/13):**

6 Let $z = \cos\theta + \mathrm{i}\sin\theta$. Use the binomial expansion of $(1+z)^n$, where n is a positive integer, to show that

$$\binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \ldots + \binom{n}{n}\cos n\theta = 2^n\cos^n\left(\tfrac{1}{2}\theta\right)\cos\left(\tfrac{1}{2}n\theta\right) - 1. \qquad [7]$$

Find

$$\binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \ldots + \binom{n}{n}\sin n\theta. \qquad [2]$$

- **Example 36:**

Prove that, for all real values of θ such that $\cos\theta \neq 0$,

$$(1 + i\tan\theta)^n = \frac{\cos n\theta + i\sin n\theta}{\cos^n\theta}.$$

In the case $n = 2p$, show by substituting a suitable value for θ in this equation that

$$1 - \binom{2p}{2} + \binom{2p}{4} - \cdots + (-1)^p\binom{2p}{2p} = 2^p\cos\left(\frac{1}{2}p\pi\right).$$